

Scoring Design

Introduction

The proposed AI evaluation system assigns quality scores to every LLM-generated response using multiple evaluation dimensions. Each dimension measures a specific aspect of response quality. These individual scores are then combined to generate an overall quality score and a human-readable explanation.

1. Scoring Dimensions

The system evaluates responses using four primary dimensions:

- **Relevance Score**
- **Accuracy Score**
- **Hallucination Score**
- **Completeness Score**

Each score ranges from **0 to 100**, where:

- **90–100:** Excellent
 - **75–89:** Good
 - **50–74:** Average
 - **Below 50:** Poor
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2. Relevance Score

Purpose

Measures how well the generated response answers the user's question.

Evaluation Criteria

- Query understanding
- Topic relevance
- Context matching

Example

Question: Explain Machine Learning.

Response: Machine Learning is a branch of AI that enables systems to learn from data.

Relevance Score: 96/100

3. Accuracy Score

Purpose

Measures the factual correctness of the generated response by comparing it with the retrieved reference documents.

Evaluation Criteria

- Correct facts
- Correct definitions
- Correct dates and names
- Consistency with retrieved documents

Example

Retrieved Document:

Python was created by Guido van Rossum.

Generated Response:

Python was created by Guido van Rossum.

Accuracy Score: 98/100

4. Hallucination Score

Purpose

Measures whether the response contains fabricated or unsupported information.

Evaluation Criteria

- Unsupported claims
- Fake facts
- Fake citations
- Faithfulness to retrieved context

Example

Retrieved Context:

Earth revolves around the Sun.

Generated Response:

Earth revolves around Mars.

Hallucination Score: 20/100

5. Completeness Score

Purpose

Measures whether all important aspects of the user's query are covered.

Evaluation Criteria

- Coverage of key concepts
- Missing information
- Level of detail

Example

Question:

Explain Machine Learning.

Response includes:

- Definition ✓
- Types ✓
- Applications ✓
- Advantages ✓

Completeness Score: 95/100

6. Overall Score Formula

The final response quality score is calculated using a weighted average of all evaluation dimensions.

Formula

```
Overall Score =  
(0.30 × Relevance Score)  
+ (0.30 × Accuracy Score)  
+ (0.20 × Hallucination Score)  
+ (0.20 × Completeness Score)
```

Example

Dimension	Score
Relevance	95
Accuracy	92
Hallucination	90
Completeness	88

Overall Score = $(0.30 \times 95) + (0.30 \times 92) + (0.20 \times 90) + (0.20 \times 88) = 91.7$

Final Rating: Excellent

7. Reason Generation

Along with the numerical score, the system generates a natural language explanation to help users understand the evaluation results.

Example

Overall Score: 92/100

Reason:

The response is highly relevant to the user's question, factually accurate, well-supported by the retrieved documents, and covers most of the important concepts. No significant hallucinations were detected.

Conclusion

The proposed scoring mechanism provides a structured and transparent method for evaluating LLM-generated responses. By combining **Relevance**, **Accuracy**, **Hallucination**, and **Completeness** scores into a

weighted overall score, the system delivers both quantitative evaluation and qualitative feedback. This design improves response reliability, simplifies performance analysis, and supports continuous improvement of the AI system.